Insect pest specifies identification

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Abstract *— This project investigates insect pest classification on IP102, a 102-class dataset with a strongly long-tailed distribution. Three approaches were evaluated: a baseline ResNet-50, an imbalance-aware ResNet-50, and EfficientNet-B3. The imbalance-aware model combined resampling, stronger minority-class augmentation, and class-weighted loss. Beyond overall performance, we examined whether minority classes are necessarily difficult and whether errors concentrate among mutually confusing classes. Per-class metrics and a weighted confusion graph identified seven high-confidence clusters containing 21 classes and separated cluster-hard, diffuse-hard, easy, and uncertain groups. The three approaches achieved broadly similar results, and imbalance-aware training provided no clear overall advantage. Minority classes were not consistently difficult. The cluster-hard group achieved a validation macro F1-score of 0.534, while diffuse-hard achieved 0.466, showing that poor performance arises from both concentrated inter-class confusion and diffuse class-specific difficulty. Therefore, class imbalance alone does not fully explain recognition difficulty on IP102. This project's git repo is available at: https://github.com/TmsJS/COMP9444_26T2_GroupProject.git*

Keywords *— Insect pest classification on dataset IP102, ResNet-50, Efficient Net, class imbalance analysis, confusion matrix with graph analysis*

I. Introduction

Pest identification is important for precision agriculture, crop protection, pest monitoring, and ecological research. Manual identification is time-consuming and relies heavily on specialised knowledge. Deep learning provides a more efficient approach to automatic pest identification [1].

This project uses IP102, which contains 75,222 images from 102 insect classes [2]. The task is challenging because related species may have similar visual features, while images within one class vary in pose, scale, lighting, and background. IP102 also has a strongly long-tailed distribution, with large differences in sample frequency between classes [3].

We compared a baseline ResNet-50 [4], an imbalance-aware ResNet-50, and EfficientNet-B3 [5]. The imbalance-aware model combined resampling, stronger minority-class augmentation, and class-weighted loss. Performance was evaluated using accuracy, macro F1-score, G-mean, and per-class metrics.

In addition to comparing models, we investigated whether minority classes are always difficult and which classes are repeatedly confused. The validation confusion matrix was represented as a weighted undirected graph, where edge weights measured reciprocal confusion. Graph-based community detection was used to identify confusion clusters, while the remaining classes were divided into easy, diffuse-hard, and uncertain groups. This analysis distinguishes concentrated inter-class confusion from more diffuse class-specific difficulty and shows that class imbalance alone cannot fully explain classification performance.

II. LITERATURE REVIEW

Convolutional neural networks can learn visual features directly from images and have become widely used for pest identification. Wu et al. [2] introduced IP102, which contains more than 75,000 images from 102 pest categories and follows a naturally long-tailed distribution. Their experiments identified inter-class similarity, intra-class variation, and class imbalance as major challenges. He et al. [4] introduced residual networks, whose shortcut connections support the training of deeper models. Building on this architecture, Ren et al. [1] proposed a feature-reuse residual network to strengthen pest feature representation, while Liu et al. [6] developed DFF-ResNet to improve residual feature extraction. Nanni et al. [7] also explored CNNs combined with bio-inspired and saliency-based features for insect pest recognition.

Alternative architectures have also been investigated. Peng and Wang [8] combined convolutional and Transformer components for insect pest classification. Tan and Le [5] proposed EfficientNet, which jointly scales network depth, width, and input resolution to balance predictive performance and computational cost. EfficientNet-B3 therefore provides a suitable alternative for comparison with ResNet-50.

Class imbalance remains a major challenge in long-tailed recognition. Zhang et al. [3] observed that models tend to favour majority classes, reducing performance on minority classes. Common approaches include resampling, data augmentation, and loss reweighting. Kang et al. [9] showed that representation and classifier learning can be decoupled and that classifier rebalancing can improve long-tailed recognition. Yang et al. [10] further reviewed long-tailed visual recognition methods and highlighted the limitations of relying on overall accuracy when class frequencies differ substantially.

However, class frequency alone may not explain why particular pest classes remain difficult. Errors may also result from visual similarity between species or other class-specific characteristics. Therefore, this project compares baseline ResNet-50, imbalance-aware ResNet-50, and EfficientNet-B3 using both overall and per-class metrics. It further analyses reciprocal classification errors as a weighted graph to identify confusion clusters and distinguish concentrated inter-class confusion from diffuse class-specific difficulty.

III. METHODS

Three classification approaches were evaluated: a baseline ResNet-50, an imbalance-aware ResNet-50, and EfficientNet-B3. All models were initialised with ImageNet-pretrained weights, adapted with a 102-class classification layer, and fine-tuned on IP102. The baseline ResNet-50 provided the reference performance, while EfficientNet-B3 was included to examine whether an alternative CNN architecture improved classification performance.

The imbalance-aware ResNet-50 retained the baseline architecture but modified the training strategy. Minority classes were oversampled and received stronger online augmentation, while highly represented classes were undersampled. Class-weighted cross-entropy was also applied to increase the contribution of underrepresented

classes. Keeping the architecture unchanged allowed a controlled comparison between standard and imbalance-aware training.
To investigate class-specific difficulty, the baseline ResNet-50 validation confusion matrix was represented as a weighted undirected graph. Each node represented an insect class, and each edge weight measured reciprocal confusion between two classes. Louvain community detection was used to identify high-confidence confusion clusters. Based on cluster membership and validation performance, the 102 classes were subsequently divided into cluster-hard, easy, diffuse-hard, and uncertain groups.

IV. EXPERIMENTS

A. Dataset and Data Exploration

The experiments used the IP102 insect pest dataset, which contains 75,222 images from 102 insect pest classes. The official split includes 45,095 training images, 7,508 validation images, and 22,619 test images. These original splits were retained for all experiments so that the three classification approaches were evaluated under the same data partition.

Analysis of the training set revealed a strong long-tailed class distribution. A small number of classes contained substantially more samples than the remaining classes. The largest training class contained 3,444 images, while the smallest contained only 42, producing an imbalance ratio of approximately 82:1. As shown in Figure. 1, class frequencies decrease sharply, with many classes containing only a few hundred training samples or fewer. This distribution motivated the introduction of an imbalance-aware ResNet-50 training strategy.

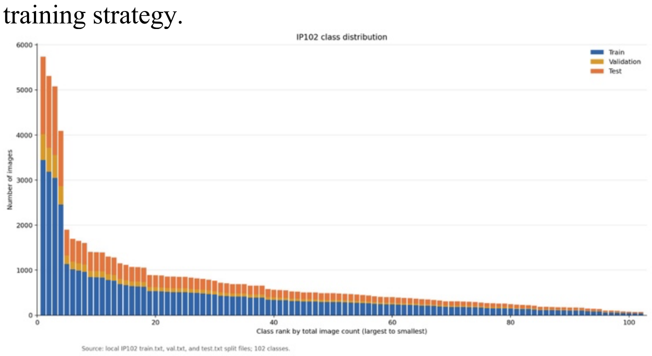

Figure 1. IP102 class distribution

The long-tailed distribution shown in *Figure. 1* also affects how model performance should be interpreted. Overall accuracy tends to place greater emphasis on well-represented classes, so strong accuracy does not necessarily indicate good recognition of minority classes. For this reason, Macro-F1 and G-mean were used alongside accuracy to provide a more balanced evaluation across all 102 classes.

The validation and test sets were left unchanged throughout the experiments. No over-sampling, under-sampling, or random augmentation was applied during evaluation. This ensured that all models were tested on the original data distribution and under consistent conditions.

B. Training Setup

For the baseline ResNet-50, all input images were resized to 224 × 224 pixels. Random horizontal flipping was applied during training, followed by ImageNet normalization. The model was trained with a batch size of 64 for a maximum of 40 epochs. Standard cross-entropy loss was used with stochastic gradient descent. The initial learning rate was set to 0.01, with momentum of 0.9 and weight decay of 0.0005. A dropout rate of 0.3 was applied before the final classification layer.

EfficientNet-B3 used an input resolution of 300 × 300 pixels and a batch size of 16. The remaining optimisation settings were kept consistent with the baseline ResNet-50, including SGD with an initial learning rate of 0.01, momentum of 0.9, and weight decay of 0.0005. Random horizontal flipping and ImageNet normalization were also used during training. The model was trained for a maximum of 40 epochs.

For all models, the learning rate was reduced when validation Macro-F1 failed to improve for three consecutive epochs. Early stopping was triggered after six epochs without improvement. The checkpoint with the highest validation Macro-F1 was retained for final evaluation.

C. Imbalance-Aware Training

The imbalance-aware ResNet-50 retained the main architecture and optimisation settings of the baseline model while modifying the training strategy to reduce the effect of class imbalance. Classes containing fewer than 150 training samples were oversampled to approximately 150 samples per epoch. Classes containing more than 2,500 samples were under sampled to approximately 2,500 samples per epoch. Classes between these thresholds retained their original sampling frequency.

Stronger online augmentation was also applied to minority classes. This included random resized cropping, horizontal flipping, rotation, and mild colour jitter. Classes outside the minority group continued to use the baseline horizontal-flip augmentation.

Weighted cross-entropy loss was also introduced. Class weights were calculated from the original training frequencies using an inverse-square-root weighting scheme. This assigned greater importance to less represented classes while limiting the influence of highly represented classes during optimisation.

D. Performance Evaluation

Model performance was evaluated using accuracy, Macro-F1, and G-mean. Macro-F1 assigns equal importance to each of the 102 classes and was therefore selected as the main validation metric for model selection, learning-rate scheduling, and early stopping. G-mean, calculated from the per-class recall values, was included to assess whether the models maintained balanced recognition performance across all classes and to penalise extremely poor recall in individual classes.

Accuracy measures the overall proportion of correctly classified test images and provides a direct measure of general classification performance. However, because IP102 has a highly imbalanced class distribution, accuracy is influenced more strongly by classes containing large numbers of samples and therefore does not fully describe performance across all classes. Macro-F1 addresses this limitation by assigning equal weight to every class, while G-mean is particularly sensitive to classes with very low recall.

In addition to the overall metrics, precision, recall, F1-score, and support were calculated separately for each of the 102 classes. The per-class results were used to identify the insect classes on which ResNet-50 performed well or poorly and to determine whether classification difficulty was concentrated in particular classes.

For a more detailed per-class analysis, the 102 classes were divided into four performance groups according to their F1-scores: poor ($F1 < 0.50$), moderate ($0.50 \leq F1 < 0.70$), good ($0.70 \leq F1 < 0.85$), and excellent ($F1 \geq 0.85$). The number of classes in each group was recorded, and representative classes from different performance levels were examined using their precision, recall, F1-score, and support.

A separate minority-class inspection was also conducted to examine whether minority classes were consistently more difficult to classify. Class support was considered together with per-class F1-score, and representative minority and non-minority classes with different performance levels were compared. The average support of the poor and moderate performance groups was also examined. This analysis was used to investigate the relationship between class frequency and classification performance and to determine whether minority status alone could explain poor recognition performance.

E. *Confusion-Cluster Analysis*

The baseline ResNet-50 validation confusion matrix was converted into a weighted undirected graph. Each node represented a class, and each edge represented reciprocal confusion. An edge was retained when each direction contained at least three errors, the combined count was at least ten, and the harmonic mean of the two directional error rates was at least 0.05. Louvain community detection was applied with resolution 1.0 and seed 0.

Communities with a visual-confusion score of at least 50 were retained. Their members were classified as *cluster-hard*. Non-members were classified as easy when support was at least 20, F1-score and recall were at least 0.70, and the dominant wrong prediction represented less than 15% of support. Classes with F1-score or recall below 0.60 were classified as *diffuse-hard*; the remaining classes were classified as uncertain.

V. RESULTS

Table. 1 and *Figure. 2* summarise the final test performance of the baseline ResNet-50, imbalance-aware ResNet-50, and EfficientNet-B3 on the IP102 dataset. All three models achieved similar overall accuracy, ranging from 71.88% to 72.17%. EfficientNet-B3 achieved the highest accuracy at 72.17%, followed by the imbalance-aware ResNet-50 at 71.98% and the baseline ResNet-50 at 71.88%.

Model	Accuracy	Macro-Precision	Macro-Recall	Macro-F1	G-Mean
Baseline ResNet50	0.7188	0.6645	0.6351	0.6456	0.6034
Imbalance-aware	0.7198	0.6862	0.6160	0.6347	0.5681
EfficientNet-B3	0.7217	0.6555	0.6456	0.6455	0.6093

Table 1. Overall Test Performance of Three Classification Models

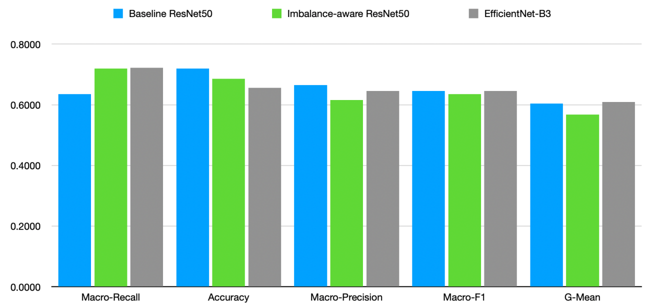

Figure 2. Comparison of Test Performance Metrics across the Three Classification Models

The class-balanced metrics showed clearer differences between the three models. The imbalance-aware ResNet-50 achieved the highest Macro-Precision, but its Macro-Recall decreased from 0.6351 to 0.6160 compared with the baseline ResNet-50. Macro-F1 also decreased from 0.6456 to 0.6347, while G-mean fell from 0.6034 to 0.5681. These results indicate that the imbalance-aware training strategy did not improve performance consistently across the 102 classes.

EfficientNet-B3 achieved the highest Macro-Recall and G-mean, together with the highest overall accuracy. Its Macro-F1 score of 0.6455 was almost identical to the baseline ResNet-50 score of 0.6456. This indicates that EfficientNet-B3 provided slightly better recall balance across classes, while the baseline ResNet-50 remained competitive with a simpler training configuration.

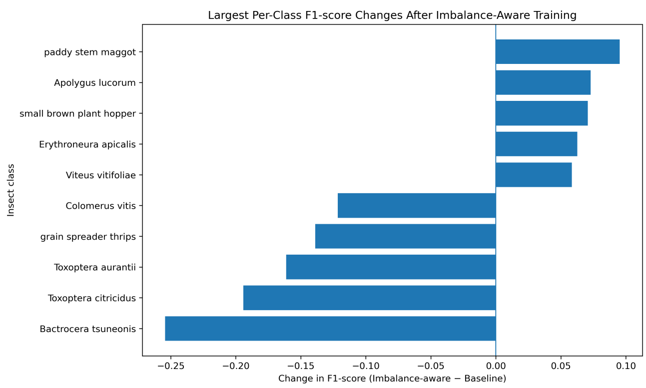

Figure 3. Largest Per-Class F1-score Changes After Imbalance-aware Training

The per-class results provide further insight into the effect of imbalance-aware training. *Figure. 3* presents classes with the largest increases and decreases in F1-score when comparing the imbalance-aware ResNet-50 with baseline model.

Several classes benefited from the imbalance-aware strategy. For example, the F1-score of *paddy stem maggot* increased from 0.41 to 0.50, while *small brown plant hopper* improved from 0.44 to 0.51. These gains suggest that increased sampling, stronger minority-class augmentation, and weighted loss helped selected classes. The same strategy also reduced performance for other classes. For example, the F1-score of *rice leaf roller* decreased from approximately 0.75 to 0.71, with larger reductions observed for several other classes. The gains were therefore class-dependent rather than consistent across dataset, which helps explain lower overall Macro-F1 and G-mean of the imbalance-aware model.

No.	label	insect_name	precision	recall	f1-score	support
8	29	bird cherry-oataphid	0.3884	0.4085	0.3982	213
10	54	Thrips	0.4495	0.3712	0.4066	264
39	39	beet army worm	0.5771	0.6058	0.5911	482
73	24	aphids	0.7405	0.7917	0.7652	1229
87	78	Ceroplastes rubens	0.825	0.8462	**0.8354**	**78**
88	61	Brevipoalpus lewisi McGregor	0.9048	0.7917	**0.8444**	**24**
100	73	Papilio xuthus	0.9764	0.9185	0.9466	135
101	62	oides decempunctata	0.9524	0.9412	**0.9467**	**85**

Table 2. Selected Per-Class Results across Different Support and F1-Score Levels

The per-class metric results from *Table. 2* show that minority status alone did not determine classification difficulty. Although the average support in the poor-performance group increased from 124 to 159 in the moderate group, some minority classes achieved high F1-scores, such as *Brevipoalpus lewisi McGregor* (support 24, F1 = 0.8444) and *oides decempunctata* (support 85, F1 = 0.9467). Conversely, *Thrips* achieved only 0.4066 despite having a support of 264.

cluster_name	classes	class_members
cluster_001	3	legume **blister beetle** | **blister beetle** | lytta polita
cluster_002	3	black **cutworm** | large **cutworm** | yellow **cutworm**
cluster_003	3	brown **plant hopper** | white backed **plant hopper** | **small** brown **plant hopper**
cluster_004	2	red spider | Panonchus citri McGregor
cluster_005	3	english grain aphid | green bug | bird-cherry-oataphid
cluster_006	3	Miridae | alfalfa **plant bug** | tarnished **plant bug**
cluster_007	4	**army worm** | cabbage **army worm** | beet **army worm** | Prodenia litura

Table 3. Visual Confusion Clusters Identified from the Validation Set

split	evaluation_level	classes	samples	accuracy	precision	recall	macro_f1	g_mean
test	original	102	22619	**0.72**	0.66	0.64	**0.65**	0.60
test	merged	88	22619	**0.78**	0.71	0.67	**0.69**	0.65

Table 4. Test Performance before and after Confusion-Based Class Merging

The confusion-cluster analysis result from *Table. 3* identified several visually similar insect classes that were frequently misclassified as one another. After merging 21 classes into seven confusion-based groups, the number of evaluation classes was reduced from 102 to 88. *Table. 4* shows that re-evaluation on the same test set increased accuracy from 0.72 to 0.78 and Macro-F1 from 0.65 to 0.69. This improvement does not indicate a change in the model itself; rather, it shows that a substantial proportion of errors occurred between visually similar classes within the identified clusters.

VI. CONCLUSION

This project evaluated baseline ResNet-50, imbalance-aware ResNet-50, and EfficientNet-B3 for fine-grained insect pest classification on the IP102 dataset. All three models achieved similar test accuracy of approximately 72%. EfficientNet-B3 achieved the highest accuracy, Macro-Recall, and G-mean, while the baseline ResNet-50 achieved a nearly identical Macro-F1 using a simpler training configuration. The imbalance-aware ResNet-50 improved several individual classes and achieved the highest Macro-Precision, but its lower Macro-Recall, Macro-F1, and G-mean did not show that combined resampling, augmentation, and loss-weighting strategy produce a consistent improvement across all classes.

The per-class analysis further showed that minority status alone did not determine classification difficulty. Some minority classes achieved high F1-scores, whereas several more frequently represented classes remained difficult to recognise. The graph-based confusion analysis identified seven clusters containing 21 classes that were frequently confused with one another. After these classes were merged into seven confusion-based groups, the evaluation space was reduced from 102 to 88 classes, increasing accuracy from 0.72 to 0.78 and Macro-F1 from 0.65 to 0.69. This does not represent an improvement in the original model, but demonstrates that a substantial proportion of its errors occurred between visually similar classes within the identified clusters.

Overall, these findings indicate further improvement requires class-specific analysis rather than a single rebalancing strategy. Future work should assess whether *cluster-hard* categories are visually distinguishable and represent reasonable classification targets, then consider hierarchical classification or validated label merging. *Diffuse-hard* classes should be examined for image quality, object size, background complexity, within-class variation, and label noise. Controlled ablation experiments should also isolate the effects of resampling, augmentation, and weighted loss.

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